

EARTH WHISPER

EARTH EVENT CASE FILE

Sandhya River near Babuganj, Barishal, Bangladesh

22.4900 N | 90.1850 E

AOI CENTER

Investigation period: June-September 2026

Case File ID: Sandhya Investigation

Verified evidence dossier

NISAR | Sentinel-1 | HLS | GPM IMERG | comparison-aware anomaly analysis

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Executive Summary

LOCATION Sandhya River near Babuganj	AOI CENTER 22.4900 N / 90.1850 E
INVESTIGATION PERIOD June-September 2026	PRIMARY OBSERVATION NISAR interferometric radar
SUPPORTING RADAR Sentinel-1	OPTICAL VERIFICATION Tested - inconclusive / many not tested
ENVIRONMENTAL CONTEXT GPM IMERG rainfall	GROUND TRUTH Not available

SCIENTIFIC FINDING

Earth Whisper detected localized changes in NISAR radar coherence within the Sandhya AOI and used Sentinel-1, temporal observations, rainfall context, optical screening, and statistical anomaly analysis to prioritize candidate regions for investigation. The evidence identifies radar-signal anomalies, but does not by itself establish a specific physical cause such as erosion, flooding, or bank migration.

Evidence language

This case file separates direct observations, independent radar cross-checks, environmental context, statistical unusualness, and missing or inconclusive evidence. The ML stage prioritizes unusual feature patterns; it is not a physical-cause classifier or an AI prediction.

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Investigation Location and Exact AOI

AOI Sandhya River near Babuganj	CENTER 22.4900 N / 90.1850 E
UTM EPSG:32646	RETENTION 95.822736%

Geometry

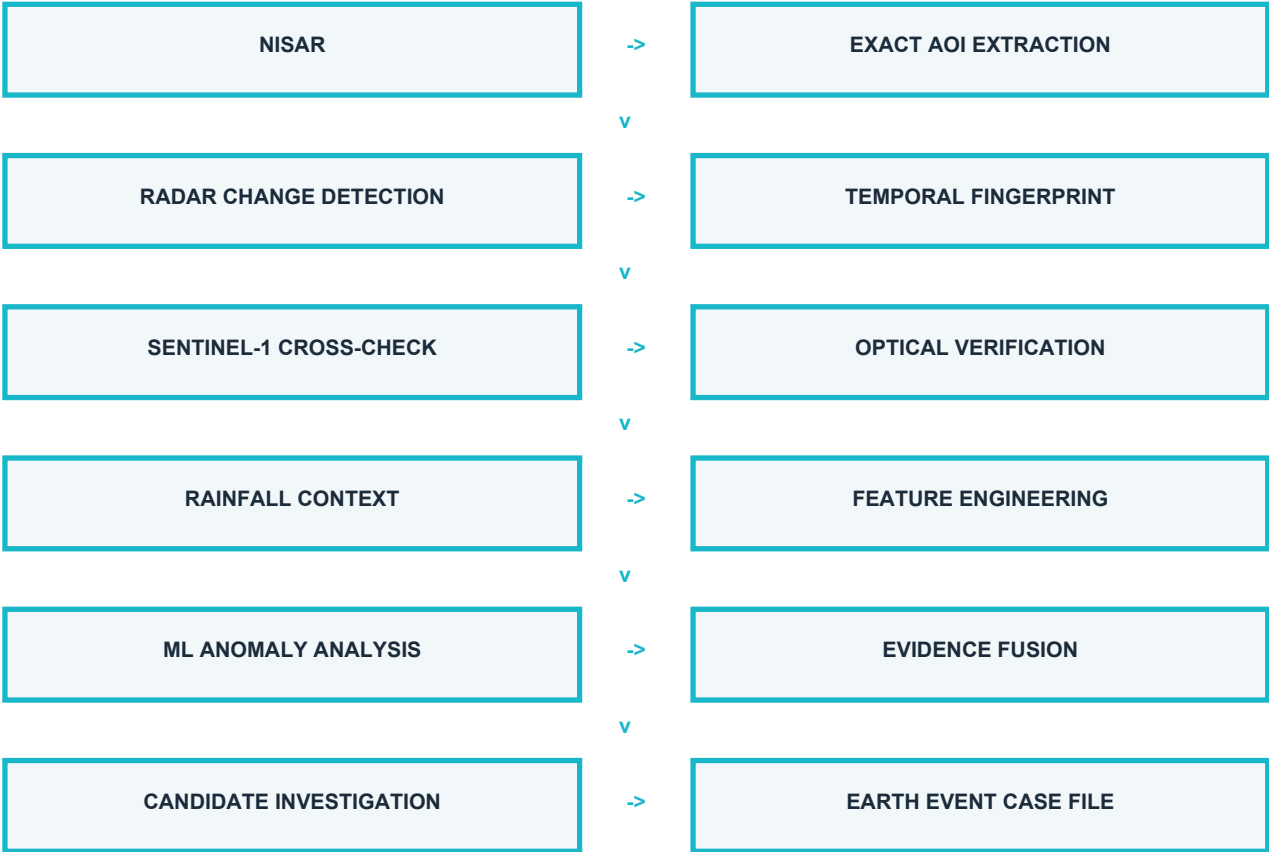
The investigation used a fixed AOI around the Sandhya River near Babuganj, Barishal. Candidate-region statistics were calculated with the exact AOI mask rather than treating the full rectangular extraction as the AOI.

Field	Verified value
AOI WKT	POLYGON((90.15 22.47,90.22 22.47,90.22 22.51,
Bounds	90.15 to 90.22 E; 22.47 to 22.51 N
Extraction window	Columns 5584:5948; rows 9928:10157
Rectangular window	364 x 229 pixels = 83,356 pixels
Exact AOI retained	79,874 pixels
Outside exact AOI	3,482 pixels

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Investigation Pipeline

Earth Whisper is a multi-stage evidence workflow. The case file is not a direct NISAR-to-AI answer: each later stage consumes outputs from the earlier verified processing stage.



INTERPRETATION BOUNDARY

Deterministic scientific processing produced the measurements. The comparison-aware ML stage identified statistically unusual candidate feature patterns, and the AI Investigator organized verified evidence for review. Neither stage establishes a physical event by itself.

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NISAR Dataset and Observations

GUNW products provide the primary interferometric coherence observations used in this investigation. GSLC supports radar amplitude/backscatter characterization, and GCOV is available as a NISAR surface/backscatter/polarimetric product. The products are not treated as interchangeable or equally decisive in the final evidence statement.

Product	Role in this investigation
GUNW	Primary interferometric evidence for coherenc
GSLC	Radar amplitude/backscatter characterization.
GCOV	NISAR surface/backscatter/polarimetric data p

Four key GUNW observations

Product	Dates	Track / frame	Orbit	Polarization
GUNW 023 091	20 Jun -> 02 Jul 2026	91 / 77	Descending	HH
GUNW 025 091	14 Jul -> 31 Aug 2026	91 / 77	Descending	VV
GUNW 026 091	26 Jul -> 19 Aug 2026	91 / 77	Descending	VV
GUNW 029 091	31 Aug -> 12 Sep 2026	91 / 77	Descending	HH

SCIENTIFIC NOTE

These are separate interferometric observation pairs with different temporal baselines and polarizations. They are not one uniform, continuous same-polarization coherence time series. The 025 and 029 observations share 31 Aug as a connected acquisition date, but the full set remains a grouped evidence fingerprint.

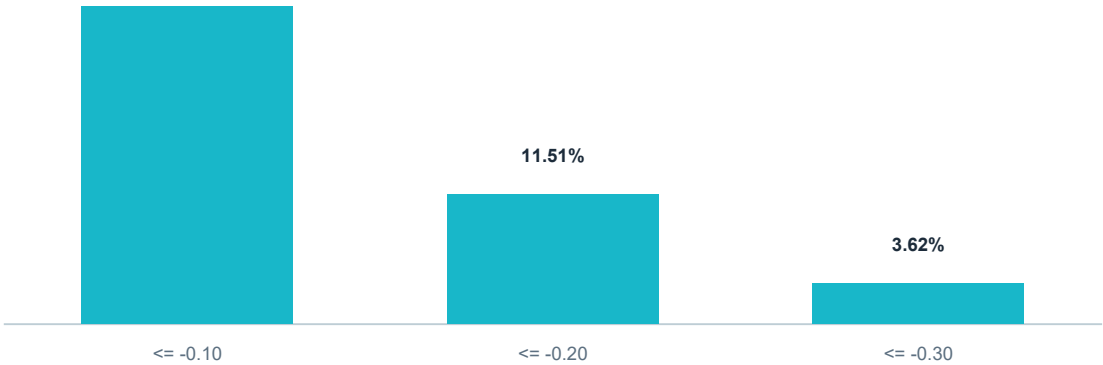
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NISAR Coherence Change - HH 023 -> 029

HH 023 -> 029

VALID PIXELS 79,874	MEAN -0.013475091
MEDIAN -0.011641301	P05 / P95 -0.27407 / 0.24227

Statistic	Value	Statistic	Value
P25	-0.11565859	P75	0.08972979
Pixels <= -0.10	28.2169%	Pixels <= -0.20	11.5069%
Pixels <= -0.30	3.6207%	Source	Exact AOI NPY output



The HH comparison shows a negative mean and median across the exact AOI. The distribution is reported separately from VV and is not combined with it.

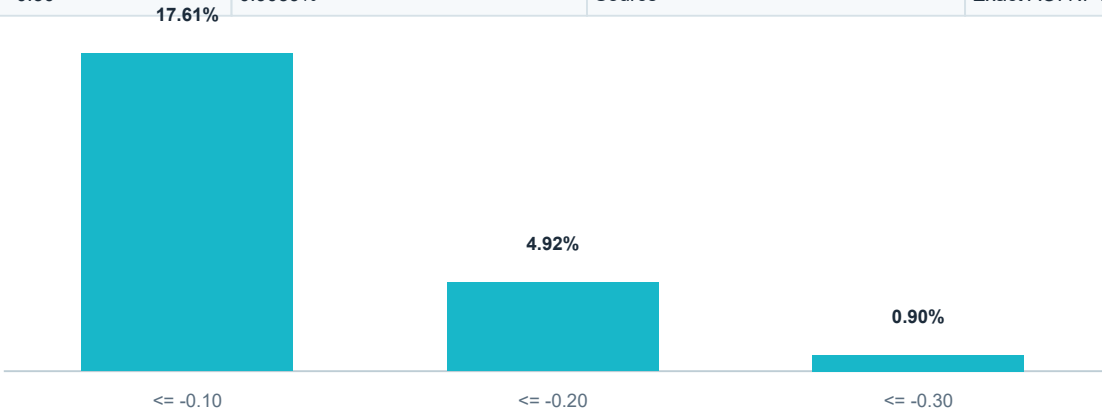
05

NISAR Coherence Change - VV 025 -> 026

VV 025 -> 026

VALID PIXELS 79,874	MEAN 0.025675088
MEDIAN 0.023278385	P05 / P95 -0.19891 / 0.25648

Statistic	Value	Statistic	Value
P25	-0.06578035	P75	0.11580434
Pixels <= -0.10	17.6140%	Pixels <= -0.20	4.9228%
Pixels <= -0.30	0.9039%	Source	Exact AOI NPY output

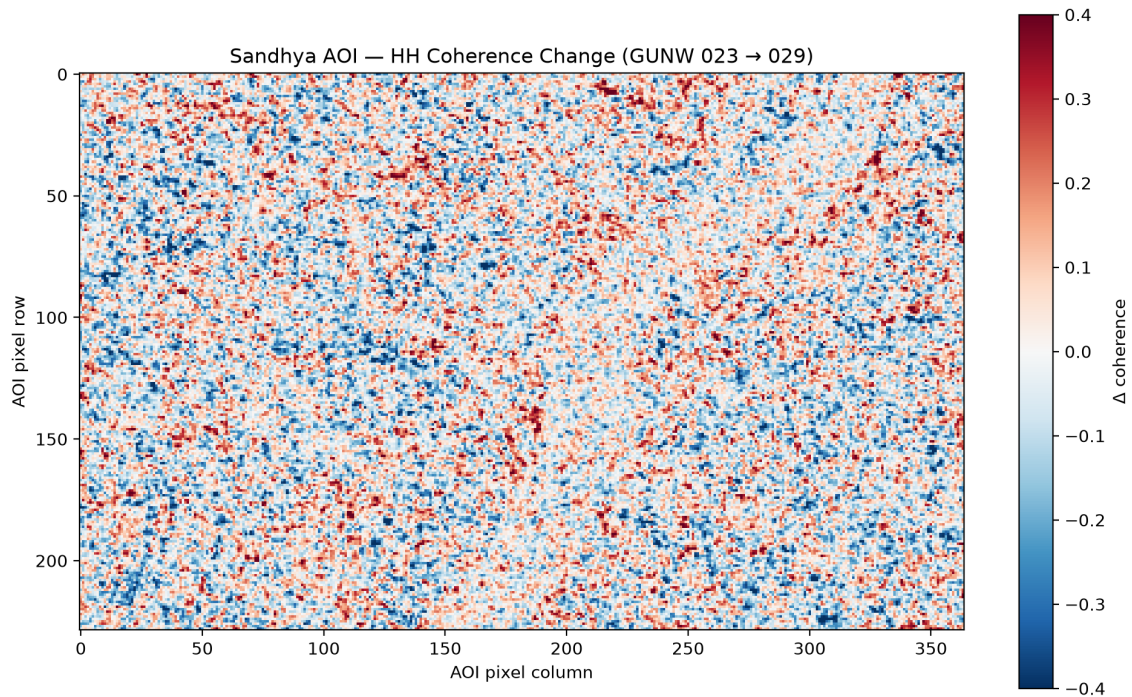


The VV comparison has a positive AOI mean and median while still containing a lower-tail population of negative changes. HH and VV distributions are reported separately and are not combined.

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Verified NISAR Figures

Figure 1 - NISAR HH coherence-change map



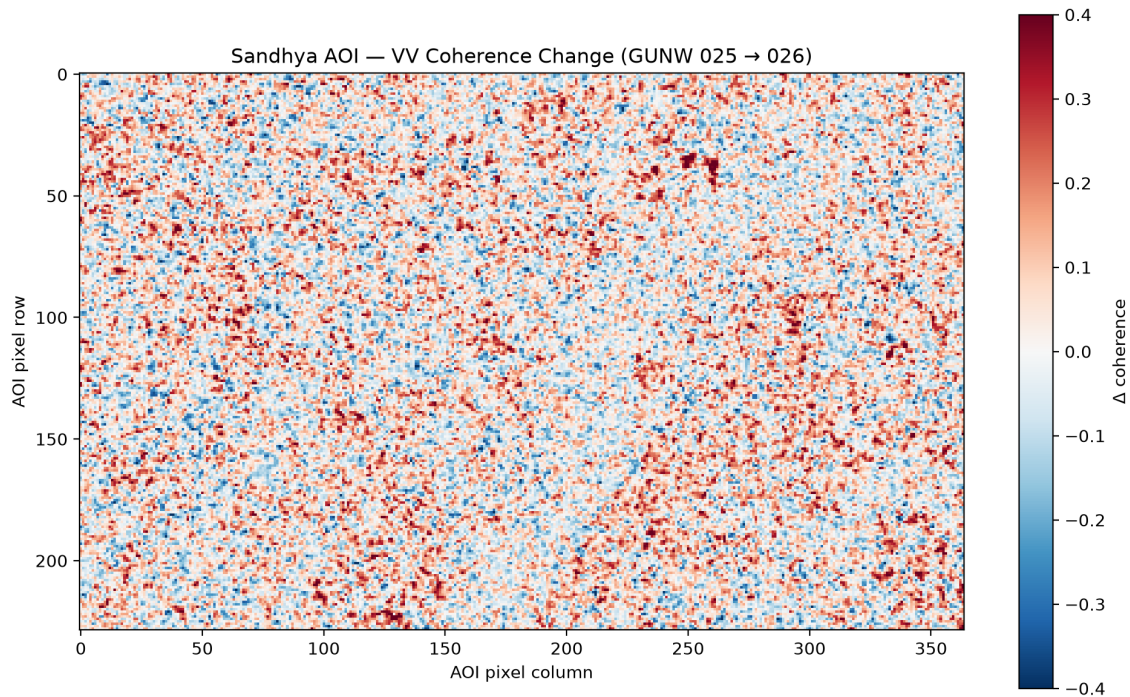
Source: Earth Whisper Step 2 - verified exact-AOI output

20 Jun 2026 -> 02 Jul 2026 / 31 Aug 2026 -> 12 Sep 2026; HH polarization. Negative values indicate lower coherence in the later observation under the comparison. This is a radar-signal change indicator, not proof of a specific physical process.

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Verified NISAR Figures - Continued

Figure 2 - NISAR VV coherence-change map



Source: Earth Whisper Step 2 - verified exact-AOI output

14 Jul 2026 -> 31 Aug 2026 / 26 Jul 2026 -> 19 Aug 2026; VV polarization. Negative values indicate lower coherence in the later observation under the comparison. This is a radar-signal change indicator, not proof of a specific physical process.

FIGURE CAVEAT

The maps show derived coherence-change indicators within the Sandhya investigation area. They must not be labeled as erosion maps, landslide maps, or flood maps without independent evidence that establishes such a physical cause.

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Candidate Region Detection

Candidate generation used a NISAR coherence-change threshold of delta coherence ≤ -0.20 , a minimum connected region of 10 pixels, 8-connected components, and a 400 m2 pixel area. The resulting 211 records represent spatially connected threshold-exceeding regions, not confirmed events.

CANDIDATE RECORDS 211	HH COMPARISON 195
VV COMPARISON 16	PIXEL AREA 400 m2
CONNECTIVITY 8-connected	THRESHOLD delta coherence ≤ -0.20

Candidate interpretation

SELECTION BOUNDARY The candidate pool was generated from NISAR change detection and is not an unbiased sample of every location in the AOI. Later Sentinel-1, optical, rainfall, and statistical stages evaluate or contextualize this selected pool.

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Temporal Fingerprint

The temporal fingerprint uses four separate interferometric pairs. For the selected investigation set, coherence values are preserved in the machine-readable case file and in the candidate records below. The pairs are shown as grouped observations rather than one connected time-series line.

Pair	Dates	Polarization	Interpretation
01	20 Jun -> 02 Jul 2026	HH	Separate interferometric pair
02	14 Jul -> 31 Aug 2026	VV	Separate interferometric pair
03	26 Jul -> 19 Aug 2026	VV	Separate interferometric pair
04	31 Aug -> 12 Sep 2026	HH	Separate interferometric pair

Representative investigated candidates

Region	Pol	Area	Verified NISAR measurements
265	HH	5200 m2	{'GUNW 023 (HH) coherence median': '0.6176549
1786	HH	4400 m2	{'GUNW 023 (HH) coherence median': '0.6266696
296	HH	19600 m2	{'GUNW 023 (HH) coherence median': '0.6274597
478	HH	25600 m2	{'GUNW 023 (HH) coherence median': '0.5766341
167	HH	4000 m2	{'GUNW 023 (HH) coherence median': '0.5960484
246	HH	7200 m2	{'GUNW 023 (HH) coherence median': '0.5975152
188	HH	8400 m2	{'GUNW 023 (HH) coherence median': '0.5603159
970	HH	7200 m2	{'GUNW 023 (HH) coherence median': '0.5527639

Different interferometric pairs and polarizations are shown separately. A candidate-level technical record is retained in the JSON appendix for all 32 investigated candidates.

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Sentinel-1 Cross-check

Sentinel-1 provides an additional radar observation with a different acquisition history and measurement behavior. It is a cross-check of NISAR-selected candidate regions, not independent ground truth.

Field	Verified value
Search period	June-September 2026
Same-platform pair	S1D: 03 Jul 2026 -> 15 Jul 2026 (12 days)
Path / frame	Full Path 150 / Frame 519
Valid pixels each	1,120,000
Mean dB	-10.1987 -> -10.6597; change about -0.46 dB
Median dB	-9.1307 -> -9.5977; change about -0.47 dB
Candidate overlap	211 / 211 NISAR candidates

CROSS-CHECK MEANING

Sentinel-1 provides an additional radar view of the observed change. Candidate-level agreement can support prioritization, but it does not prove erosion, flooding, bank migration, or another physical event.

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ML and Cross-radar Consistency

Comparison-aware statistical anomaly analysis

The ML stage identifies candidate regions whose Sentinel-1 feature patterns are statistically unusual compared with other candidates in the same comparison group. It is unsupervised statistical anomaly detection, not erosion classification, event probability, physical-cause classification, ground truth, or AI confidence.

Input features	Comparison groups	k values	Selection output
s1 median db; s1 min db; s1 max db; s1 pct le	HH: 195 VV: 16	3, 5, 8	Top 20 by anomaly percentile plus stable/sens

Redundant features were excluded after feature-correlation analysis. Separate HH and VV populations are used because their distributions differ and should not be treated as one homogeneous population.

Cross-radar distribution

Category	Count
Dual-radar upper-quartile stable	11
Dual-radar upper-quartile but sensitive	5
NISAR upper-quartile; S1 not upper-quartile	38
S1 upper-quartile; NISAR not upper-quartile	34
S1 upper-quartile but sensitive	6
Neither radar measurement upper-quartile	117

STATISTICAL BOUNDARY

These categories describe statistical consistency within comparison populations. They are not physical-event probabilities and should not be converted into a confidence percentage.

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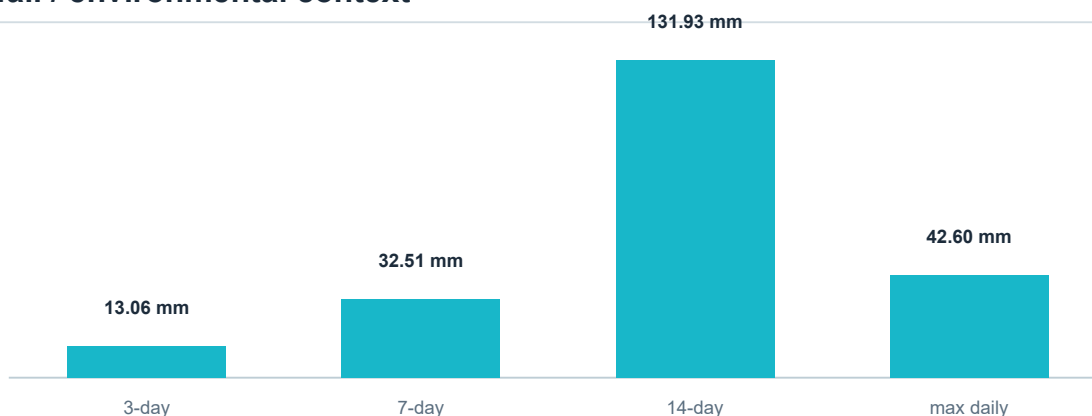
Optical Verification and Rainfall Context

Optical verification

Field	Verified value
Dataset	HLS S30 v2.0
Tile	T46QBK
Selected dates	26 Jun 2026; 14 Sep 2026
Screened candidates	6
Status	TESTED - INCONCLUSIVE
Not screened	205 candidates
Valid optical pixels	0 for the inconclusive tested cases

Optical verification was inconclusive because valid optical coverage was severely limited by high aerosol conditions. Candidates not screened remain NOT TESTED. No NDVI, NDWI, MNDWI, water extent, or optical confirmation is fabricated in this report.

Rainfall / environmental context



Context	Value
Dataset	NASA GPM IMERG Late Run V07 / GPM 3IMERGDL
12 Sep 2026, 3-day	13.056936 mm
12 Sep 2026, 7-day	32.511207 mm
12 Sep 2026, 14-day	131.931329 mm
Maximum daily	42.598942 mm
Wet days > 5 mm	9

Rainfall is environmental context and does not establish that rainfall caused the radar anomaly. The 14-day context for GUNW 023 is unavailable. For GUNW 025, the 31 Aug rainfall is end-date context for the 14 Jul -> 31 Aug interval, not rainfall across the full 48-day interval.

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AI Investigator and Investigated Candidates

The AI Investigator interpreted verified evidence that had already been produced by deterministic processing and ML analysis. It did not discover or recalculate the original satellite signal. The selection included the top 20 candidates by comparison-aware anomaly percentile plus all candidates in stable or sensitive dual-radar upper-quartile categories. After deduplication: 32 investigated candidates, 29 HH and 3 VV.

Order	Pol	Region	Area m2	Location	NISAR	S1 median	ML percentile
1	HH	265	5200	22.5058, 90.2193	-0.344	-6.78	100.0%
2	HH	1786	4400	22.4791, 90.1647	-0.308	-7.88	99.1%
3	HH	296	19600	22.5040, 90.1965	-0.308	-4.00	99.0%
4	HH	478	25600	22.4993, 90.1546	-0.293	-0.37	98.6%
5	HH	167	4000	22.5062, 90.1648	-0.298	-3.85	98.1%
6	HH	246	7200	22.5043, 90.1516	-0.323	-6.27	97.4%
7	HH	188	8400	22.5056, 90.1662	-0.336	-5.09	96.8%
8	HH	970	7200	22.4930, 90.1977	-0.273	-5.62	96.1%
9	HH	224	5600	22.5062, 90.2041	-0.298	-2.72	95.7%
10	HH	2020	4000	22.4759, 90.2118	-0.273	0.47	95.0%
11	HH	900	4000	22.4942, 90.1970	-0.342	-4.63	94.9%
12	HH	1536	5200	22.4835, 90.1976	-0.259	-3.92	94.5%
13	HH	2186	10400	22.4728, 90.2035	-0.315	1.06	94.5%
14	HH	1628	4000	22.4815, 90.1615	-0.317	-0.45	93.3%
15	HH	1653	4000	22.4810, 90.1712	-0.276	-2.26	92.6%
16	HH	1933	5200	22.4761, 90.1553	-0.307	-2.88	92.0%
17	HH	53	5200	22.5084, 90.1617	-0.304	-0.67	91.8%
18	HH	608	5200	22.4993, 90.2138	-0.240	-3.51	91.5%
19	HH	1145	10000	22.4890, 90.1536	-0.341	-0.26	90.9%
20	HH	982	9600	22.4930, 90.2163	-0.348	-1.60	88.9%
21	HH	1902	4800	22.4773, 90.1744	-0.494	-2.63	88.0%
22	HH	1640	7600	22.4821, 90.2182	-0.377	1.90	87.5%
23	HH	2035	6000	22.4757, 90.2064	-0.383	-1.69	85.0%
24	HH	1980	7600	22.4750, 90.1608	-0.332	-0.73	84.8%
25	HH	2152	5200	22.4737, 90.2086	-0.361	-1.44	81.5%
26	HH	488	5200	22.5004, 90.1595	-0.337	0.21	81.2%
27	HH	1041	5600	22.4917, 90.1970	-0.326	-2.05	81.0%
28	HH	902	4000	22.4942, 90.2060	-0.326	-0.19	77.4%
29	HH	294	5200	22.5045, 90.1781	-0.402	0.97	75.0%
30	VV	207	4000	22.5057, 90.2196	-0.242	-7.05	100.0%
31	VV	87	4800	22.5082, 90.1924	-0.296	-3.13	93.8%
32	VV	415	4800	22.4999, 90.1568	-0.290	-2.38	75.0%

The complete candidate keys, feature values, cross-radar categories, optical states, rainfall context, uncertainty, and provenance are preserved in the structured JSON case file. Region numbers in this table are readable labels; the technical identity rule remains comparison + region id.

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Evidence Fusion and Candidate Case Studies

Evidence matrix

Layer	Role	Status
NISAR	Primary radar coherence evidence	Available / supporting
Sentinel-1	Additional radar observation	Available / supporting cross-check
Optical	HLS screening	Tested - inconclusive or not tested
Rainfall	Regional environmental context	Available / context only
Statistical model	Comparison-aware kNN unusualness	Prioritization / uncertain with k-sensitivity
Ground truth	Independent physical-event label	Not available

Selected candidate case studies

Region 265 - HH - 5200 m2

Location 22.50580, 90.21930. NISAR change -0.3435; Sentinel-1 median change -6.78 dB; ML anomaly percentile 100.0%; cross-radar status dual radar upper quartile but sensitive; optical status tested but inconclusive no valid pixels. Physical cause remains unresolved.

Region 1786 - HH - 4400 m2

Location 22.47909, 90.16470. NISAR change -0.3077; Sentinel-1 median change -7.88 dB; ML anomaly percentile 99.1%; cross-radar status s1 upper quartile but sensitive; optical status tested but inconclusive no valid pixels. Physical cause remains unresolved.

Region 296 - HH - 19600 m2

Location 22.50396, 90.19650. NISAR change -0.3084; Sentinel-1 median change -4.00 dB; ML anomaly percentile 99.0%; cross-radar status s1 upper quartile nisar not upper quartile; optical status not tested. Physical cause remains unresolved.

Region 478 - HH - 25600 m2

Location 22.49933, 90.15459. NISAR change -0.2931; Sentinel-1 median change -0.37 dB; ML anomaly percentile 98.6%; cross-radar status s1 upper quartile nisar not upper quartile; optical status not tested. Physical cause remains unresolved.

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Uncertainty, Limitations, and Final Finding

Major limitations

- NISAR GUNW observations are interferometric measurements over specific acquisition pairs, not one continuous deformation time series.
- Temporal baselines and HH/VV polarizations differ and are not combined.
- Sentinel-1 is a cross-check sampled around NISAR-selected candidates, not independent ground truth.
- Only 6 of 211 candidates were optically tested; all tested cases were inconclusive and 205 were not tested.
- High aerosol conditions limited optical verification and no valid optical pixels were available for several windows.
- GPM IMERG rainfall is coarser than the AOI and is context only; early GUNW 023 14-day context is unavailable.
- Candidate detection depends on the change threshold and connected-region rules.
- ML anomaly values identify statistical unusualness and can vary with neighbourhood size; HH has 195 candidates and VV has 16.
- No independent ground-truth labels are available, so no physical event or calibrated physical-event probability can be confirmed.

EARTH WHISPER FINDING

Earth Whisper identified spatially localized radar-signal changes within the Sandhya investigation AOI. The strongest candidate regions were prioritized using comparison-aware statistical anomaly analysis and cross-radar evidence. Sentinel-1 provides additional radar observations, optical verification was limited or inconclusive, and rainfall provides environmental context without establishing causation. Without independent ground truth or conclusive optical evidence, the physical cause of the radar changes remains unresolved.

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Data Sources and Methodology

Official references

NISAR Data User Guide: <https://nisar-docs.asf.alaska.edu/>

NISAR GUNW Product Documentation: <https://nisar-docs.asf.alaska.edu/gunw/>

NISAR Vertex Search: <https://nisar-docs.asf.alaska.edu/vertex/>

ASF Vertex: <https://vertex-plus.asf.alaska.edu/>

NISAR Product Overview: <https://nisar-docs.asf.alaska.edu/products-overview/>

HLS / Harmonized Landsat Sentinel-2: <https://www.earthdata.nasa.gov/data/projects/hls>

NASA GPM / IMERG: <https://gpm.nasa.gov/data/imerg>

NASA Earthdata: <https://www.earthdata.nasa.gov/>

Methodology reference

Step	Readable methodology
1	NISAR coverage
2	NISAR AOI extraction and change
3	Temporal fingerprint
4	Optical verification
5	Environmental context
6	Feature engine
7	ML anomaly detection
8	Evidence fusion
9	Uncertainty
10	AI Investigator
11	Event Case File

Source traceability: Earth Whisper Step 2 exact-AOI NPY/PNG outputs; Step 3-6 feature and context tables; Step 8 comparison-aware anomaly and cross-radar outputs; Step 9 uncertainty annotations; Step 10 investigated candidate CSV/manifest; and Step 11 case-file JSON. The machine-readable appendix preserves the full source-record fields.